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AGILE & SCRUM

Agile methods and Scrum

How teams build software in small steps

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Goals

- After this session you can:
  - Explain why agile methods exist
  - Name the four agile values
  - Describe Scrum: roles, events and artifacts
  - Write a user story
  - Tell Scrum and Kanban apart

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The old way: Waterfall

```
Requirements  ->  Design  ->  Build  ->  Test  ->  Release

  (months)        (months)   (months)  (months)  (finally!)
```

- Plan everything first, then do one phase at a time
- Like water falling down steps: you can't go back up
- The customer sees the product at the very end

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What goes wrong?

Imagine ordering a birthday cake and tasting it for the first time at the party.

- The customer changes their mind (they always do)
- Feedback comes too late: mistakes are found at the end
- All the risk is saved for the last month

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Agile: small steps, fast feedback

Build a little. Show it. Learn. Repeat.

- Work in short cycles of 1-4 weeks
- After each cycle: something that works
- The customer gives feedback all the time
- Plans change when we learn something new

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The Agile Manifesto (2001)

We value...	more than...
People and talking together	processes and tools
Software that works	lots of documents
Working with the customer	arguing about the contract
Responding to change	following a plan

- The things on the right still matter. The left matters more.

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Some of the 12 principles

- Deliver working software often: weeks, not months
- Welcome changes, even late in the project
- Business people and developers work together every day
- Talking face to face is the best way to share information
- Keep it simple: don't build what nobody needs
- Stop regularly and ask: how can we work better?

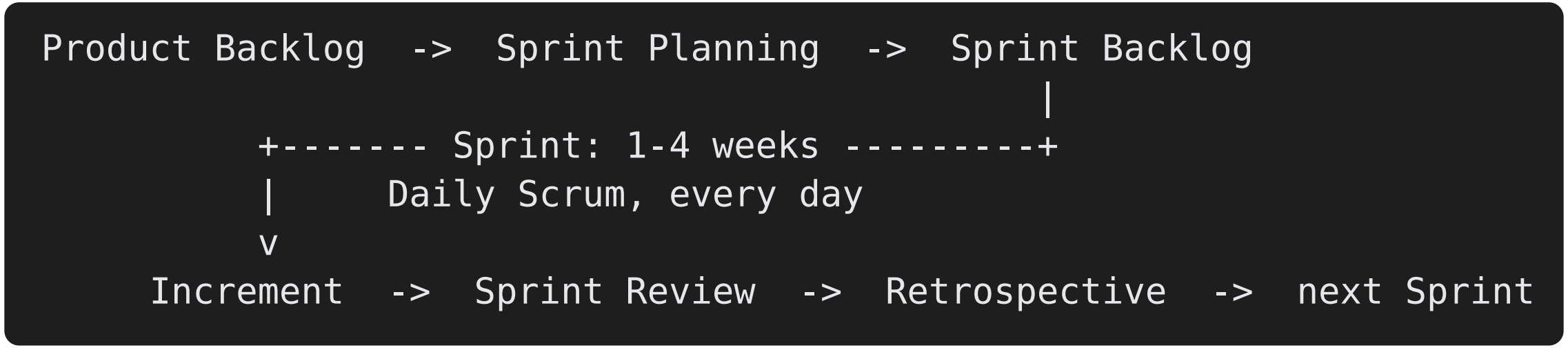
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Agile is a mindset. Methods are tools.

Method	In one sentence
Scrum	Work in sprints with clear roles and meetings
Kanban	A board with columns and a steady flow of work
XP (Extreme Programming)	Practices for code: pair programming, tests first
SAFe, LeSS	Agile for many teams in big companies

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Scrum in one picture



- 3 accountabilities (roles), 5 events, 3 artifacts

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The Scrum Team

Role	Responsible for
Product Owner	WHAT to build, and in which order. Owns the Product Backlog.
Scrum Master	HOW Scrum works. Coaches the team and removes obstacles.
Developers	Building it. Decide themselves how to do the work.

- Usually 10 people or fewer
- No boss inside the team: everyone works towards the same goal

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The Sprint

- A fixed time box: one month or less, often 2 weeks
- Has one Sprint Goal: why this sprint matters
- The goal does not change during the sprint
- A new sprint starts right after the last one ends
- All other events happen inside the sprint

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The Scrum events

Event	When	Max time*	Purpose
Sprint Planning	Start of sprint	8 hours	What can we do, and how?
Daily Scrum	Every day	15 minutes	Are we on track for the goal?
Sprint Review	End of sprint	4 hours	Show the result, get feedback
Sprint Retrospective	Very end	3 hours	How can we work better?

- * For a one-month sprint. Shorter sprints = shorter meetings.

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Daily Scrum

- 15 minutes, same time and place every day
- For the developers, to plan the next 24 hours
- Many teams ask: What did I do? What will I do? What's blocking me?
- Not a status report to a boss
- Long problems? Solve them after the meeting

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The Scrum artifacts

Artifact	What it is	Its commitment
Product Backlog	Ordered list of everything we may build	Product Goal
Sprint Backlog	What we'll do this sprint, and the plan	Sprint Goal
Increment	The new working piece of the product	Definition of Done

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User stories

```
As a <type of user>,
I want <something>,
so that <why it's useful>.

As a customer, I want to filter products by category,
so that I can find shoes faster.
```

- Written from the user's point of view
- Small enough to finish in one sprint

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Acceptance criteria and Definition of Done

Acceptance criteria	Definition of Done
For ONE story	For ALL work in the team
'Is this story doing the right thing?'	'Is the work really finished?'
Choosing 'Shoes' shows only shoes	Code reviewed by someone else
The filter works on mobile	Tests pass, merged to main

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How big is it? Story points

- Estimate size compared to other stories, not in hours
- Common scale: 1, 2, 3, 5, 8, 13
- Planning poker: everyone shows a card at the same time
- Velocity: how many points the team usually finishes per sprint

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Backlog refinement

- The Product Owner and the developers go through the backlog
- Split big items into smaller ones
- Add details and acceptance criteria
- Estimate the size
- An ongoing activity, often 1-2 hours per week

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Kanban

To do	In progress (max 3)	Review (max 2)	Done
Order history	Category filter	Login page	Search
Low stock	Shopping cart		Product page

- Visualise the work on a board
- Limit work in progress (WIP)
- No sprints: work flows continuously

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Scrum or Kanban?

	Scrum	Kanban
Rhythm	Sprints of 1-4 weeks	Continuous flow
Roles	PO, Scrum Master, Developers	No required roles
Change	Not during the sprint	Any time
Limit on work	What fits in the sprint	WIP limit per column
Good for	Building new products	Support, maintenance, operations

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Tools

- Jira: the most common in companies
- Azure DevOps: common with Microsoft tools
- Trello: simple boards, free to start
- GitHub Projects: boards right next to your code
- A whiteboard and sticky notes work too

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Common mistakes

- The Daily Scrum becomes a status report
- No retrospective, or one where nothing changes
- The Product Owner is never there
- Stories are too big to finish in a sprint
- New work is pushed in during the sprint
- 'We do Scrum, but...' (and skip the hard parts)

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Why this matters for you

- Most developer jobs work in an agile way
- Interview questions: 'What is a sprint?' 'What does a Scrum Master do?'
- As a junior: speak up at the Daily Scrum when you're stuck
- Your projects in this course can use a board and short sprints too

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Remember

- Agile = small steps, fast feedback, welcome change
- The Manifesto: 4 values, 12 principles
- Scrum: Product Owner, Scrum Master, Developers
- Events: Sprint, Planning, Daily Scrum, Review, Retrospective
- Artifacts: Product Backlog, Sprint Backlog, Increment
- Kanban: a board, WIP limits, continuous flow

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Coming up next

Git and GitHub: version control, branches and working together with pull requests.